

Quiz 1

For each of the following questions, explain your counting method. Use Durrett's $P_{n,k}$ and $C_{n,k}$ notations (and associated terminology) where it is appropriate.

1. There are five routes I drive between my house and Hood. How many different ways can I take a round trip to work and back home, if I follow a different route in each direction?

A round trip is a permutation of two routes taken from the set of five possible routes. There are $P_{5,2} = 20$ round trips.

2. A state has license plates with three letters (A through Z) and three digits (0 through 9). The digits and letters can come in any order (e.g. A99BC0 or 1Z2BB3). How many license plates like this are possible? What if the three letters and three digits cannot repeat?

Repeats possible:

We use the multiplication rule to count the license plates. We first choose 3 of the 6 slots where we will put the letters. There are $C_{6,3} = 20$ such choices. We then fill the three letter slots in order with three letters. There are $(26)(26)(26)$ ways to do that. Finally, we fill the remaining three slots in order with digits. There are $(10)(10)(10)$ ways to do that. Combining these steps with the multiplication rule, we find a total of $(20)(26)(26)(26)(10)(10)(10) = 351,520,000$ possible license plates.

No repeats:

We count the license plates without repeated letters or numbers in a similar way using the multiplication rule. We first choose 3 of the 6 slots where we will put the letters. As before, there are $C_{6,3} = 20$ such choices. We then fill the three letter slots with a permutation of 3 letters chosen from the 26 possible letters. There are $P_{26,3}$ such choices. Finally, we fill the remaining three slots with a permutation of 3 digits chosen from the 10 possible digits. There are $P_{10,3}$ such choices. Combining these steps with the multiplication rule, we find a total of $C_{6,3} P_{26,3} P_{10,3} = 224,640,000$ possible license plates without repeated letters or repeated digits.

3. Hood's men's soccer team will play 16 regular-season games. Each game can end in a win for Hood, a loss for Hood, or a tie. How many ways could Hood end up with a regular-season record of 10 wins, 4 losses, and 2 ties?

We count how many ways we can divide the 16 regular-season games into a group of 10 (the wins), a group of 4 (the losses), and a group of 2 (the ties). The multinomial counting method from Durrett tells us that there are $\frac{16!}{10!4!2!} = 120,120$ ways to do this.

We can also count the possibilities using the multiplication rule as follows: first choose 10 of the 16 games that will be wins. There are $C_{16,10}$ ways to do this. However we pick those 10 games that will be wins, there are 6 remaining games. We choose 4 of them to be losses. There are $C_{6,4}$ ways to do this. However we pick those 4 losses, there are 2 remaining games. Those games are the 2 ties. (There are $C_{2,2} = 1$ ways to pick.)

Combining the three stages with the multiplication rule, we find $C_{16,10} C_{6,4} C_{2,2} = 120,120$ possibilities, just as with the first method.